CO 2

```
public class StreamingAnalytics {

    // Segment Tree Array
    int[] segTree;
    int n;

    // Constructor
    StreamingAnalytics(int[] views) {
        n = views.length;
        segTree = new int[4 * n];
        buildTree(views, 0, n - 1, 1);
    }

    // Build Segment Tree
    void buildTree(int[] arr, int start, int end, int node) {

        if (start == end) {
            segTree[node] = arr[start];
            return;
        }

        int mid = (start + end) / 2;

        buildTree(arr, start, mid, 2 * node);
        buildTree(arr, mid + 1, end, 2 * node + 1);

        segTree[node] = segTree[2 * node] + segTree[2 *
node + 1];
```

```
    }

    // Range Sum Query
    int rangeQuery(int start, int end, int left, int right, int
node) {

        // Completely inside
        if (left <= start && end <= right)
            return segTree[node];

        // Completely outside
        if (end < left || start > right)
            return 0;

        int mid = (start + end) / 2;

        return rangeQuery(start, mid, left, right, 2 * node)
                + rangeQuery(mid + 1, end, left, right, 2 * node
+ 1);
    }

    public static void main(String[] args) {

        System.out.println("=== MEDIAVAULT STREAMING
ANALYTICS ===\n");

        System.out.println("Content Database Indexed
Successfully\n");

        // Daily Views Data
        int[] views = {1200, 1500, 1800, 2100, 1700, 2500};
```

```
        System.out.println("Daily Views:\n");

        for (int i = 0; i < views.length; i++) {
            System.out.println("Day " + (i + 1) + " : " +
views[i]);
        }

        // Create Segment Tree
        StreamingAnalytics st = new
StreamingAnalytics(views);

        // Range Query Day 2 to Day 6
        int left = 1;   // Day 2 (0-based index)
        int right = 5;  // Day 6

        System.out.println("\nRange Query:");
        System.out.println("Views From Day 2 To Day 6\n");

        int total = st.rangeQuery(0, st.n - 1, left, right, 1);

        System.out.println("Total Views = " + total);

        System.out.println("\nB+ Tree Search Completed");

        System.out.println("Segment Tree Query
Completed");

        System.out.println("\nPerformance Summary:\n");

        System.out.println("B+ Tree Search : O(log n)");
```

```
        System.out.println("Segment Tree Query : O(log
n)");

        System.out.println("\nProgram Executed
Successfully.");
    }
}
```